

EDS White Paper: Inflow & Infiltration - Measuring the Invisible Problem

Environmental Data Services

Where Quality Flows...

Executive Summary

Inflow and Infiltration (I&I) represents one of the most persistent and costly challenges facing councils and utilities. Excess stormwater and groundwater entering sewer systems not only lead to treatment inefficiencies and network overloading but also accelerate asset deterioration and increase operational costs. Beyond these operational pressures, unmanaged I&I can mask true system capacity, complicate hydraulic modelling, and distort investment planning, leading to costly over or under sizing of infrastructure projects.

Accurate measurement of I&I is **essential for making informed decisions about remediation, capacity planning, and capital investment**. In practice, this involves combining flow monitoring with rainfall analysis, network modelling, and site inspections to build a complete picture of how and where extraneous water enters the system. This white paper outlines the role of short term and longterm flow monitoring in identifying and quantifying I&I, discusses how actionable data drives targeted interventions and highlights how EDS has helped its clients reduce I&I related costs through best practice monitoring, analysis, and tailored remediation strategies.

1. Why Inflow & Infiltration is a Critical Challenge

- **Operational Impact:**
 - Overloaded sewer networks during wet weather events, often resulting in bypass pumping and emergency response operations.
 - Increased pumping and treatment costs due to unnecessary processing of stormwater, leading to higher energy consumption and reduced treatment efficiency.
 - Distorted flow data that can obscure true diurnal patterns, complicating network modelling and inflow forecasting.
- **Environmental Impact:**
 - Greater risk of **sanitary sewer overflows (SSOs)** during peak events, which can result in environmental harm, community complaints, and regulatory breaches.
 - Accelerated asset degradation due to increased hydraulic loads and the corrosive effects of fluctuating groundwater infiltration.
 - Increased greenhouse gas emissions from unnecessary pumping and treatment of extraneous flows.
- **Financial Impact:**
 - Avoidable treatment costs, particularly during peak storm events when treatment plants are already at capacity.
 - Unnecessary capital expenditure on upsizing infrastructure that could instead be addressed through I&I reduction strategies, deferring costly upgrades.
 - Higher maintenance and operational budgets required to manage increased wear and tear on pumping stations, treatment plants and conveyance systems.

Industry data suggests I&I can represent up to **50% of flow in older sewer networks** during storm events, a massive burden on system capacity, operating budgets and long term planning.

2. The Role of Flow Monitoring

Short Term Flow Monitoring

- Typically deployed for **inflow and infiltration studies**, short term monitoring is crucial when councils need rapid insights to inform network rehabilitation or planning decisions.
- Measures flow responses during wet weather events over **6–12 weeks**, capturing the dynamic impact of rainfall and storm events on sewer inflow. Data is often combined with rainfall records to perform **rainfall dependent I&I analyses**, distinguishing between inflow (rapid response) and infiltration (delayed, groundwater driven response).
- Helps identify **network subcatchments contributing to I&I**, enabling prioritisation of CCTV inspections, manhole surveys and targeted remedial work. Short term monitoring can also support model calibration by providing **high resolution data during critical storm events**.

Long Term Flow Monitoring

- Continuous monitoring provides **trend data across multiple seasons and weather conditions**, enabling utilities to assess the baseline infiltration load and its variation throughout the year.
- Identifies **groundwater infiltration** that may not be evident in short term studies, particularly in areas with seasonal groundwater fluctuations or persistent defects in aging infrastructure.
- Supports **capital works prioritisation** by quantifying the ongoing impact of I&I, allowing councils to evaluate whether capacity upgrades or targeted rehabilitation will provide the most cost effective outcomes. Long term data is also essential for measuring the **effectiveness of past interventions**.

Best Practices in Monitoring:

- **High resolution logging** (e.g., 1–5 minute intervals) for accurate event capture and diurnal pattern analysis.
 - **Pre-installation inspections** to ensure optimal site selection, assessing hydraulics, access and potential for ragging or debris accumulation.
 - **Intrinsically safe equipment** for safe deployment in live networks, complying with ATEX and IECEx standards.
 - **Integrated rainfall gauging** to correlate wet weather events with flow responses, improving the accuracy of I&I attribution.
 - **Robust QA/QC processes** including data validation, calibration checks, and remote telemetry to ensure data integrity throughout the monitoring program.
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3. Turning Data into Action

I&I data is only valuable if it leads to **targeted, cost effective decisions**.

The Pareto Principle in I&I Management

EDS's experience, consistent with findings across the water industry, shows that the **Pareto Principle often holds true**:

Approximately 20% of a sewer network typically accounts for 80% of I&I volumes.

This insight enables councils and utilities to:

- **Prioritise** rehabilitation efforts in high impact subcatchments.
- **Maximise ROI** by addressing the most significant contributors first.
- **Avoid blanket rehabilitation**, which can waste resources with minimal benefit.

From Monitoring to Action

- **Targeted Remediation:** Using data driven prioritisation to identify and repair the worst performing segments (e.g., relining, grouting).
- **Capital Works Planning:** Incorporating I&I analysis into **long term asset strategies**, reducing the need for costly upsizing projects.
- **Performance Measurement:** Post remediation monitoring confirms the **effectiveness of interventions**, validating investment decisions.

By applying this principle, EDS helps clients achieve significant reductions in I&I volumes while minimising expenditure.

4. Case Study: Regional Council I&I Reduction

Project Overview: A regional council engaged EDS to quantify I&I across its sewer network ahead of a major treatment plant upgrade. The network was experiencing frequent wet weather overflows and escalating operational costs, and the council required precise data to inform whether a multi million dollar capacity upgrade was justified or if targeted rehabilitation would be more effective.

Approach:

- EDS deployed **20 short term area velocity flow meters** across strategically chosen subcatchments, selected through pre-installation hydraulic assessments and asset condition reviews.
- **Rainfall correlation analysis** was performed using on site tipping bucket rain gauges and historical climate data to differentiate between inflow (rapid response during storms) and infiltration (sustained groundwater ingress). This allowed for clear separation of rainfall dependent inflow and base groundwater infiltration components.
- Data underwent **robust QA/QC validation**, including cross checking with SCADA pump station data and field inspections to verify anomalies, ensuring high confidence in reported volumes.

Results:

- Identified **three high priority subcatchments** contributing over **65% of I&I volume**, confirming the Pareto Principle in practice and allowing the council to focus its rehabilitation program where it would have the most impact.
- Provided data driven evidence that **avoided a \$6M treatment plant expansion**, allowing the council to redirect capital into sewer relining, manhole sealing, and inflow source reduction.
- Developed detailed **remediation prioritisation plans** with cost benefit analyses, giving the council a phased program for I&I reduction.

Outcome:

- Reduced peak wet weather flows by **35%** within two years through targeted rehabilitation measures, including structural relining and defect sealing.

- Deferred major capital expenditure on treatment upgrades and achieved significant operational cost savings in pumping and treatment.
- Established an ongoing long term monitoring program to validate the effectiveness of rehabilitation efforts and provide continuous insights for asset management planning.

Conclusion

Inflow and infiltration is an **invisible but solvable problem**. With the right monitoring program, actionable insights, and a structured approach to prioritising investments, councils and utilities can **reduce operational costs, defer unnecessary capital works, and improve network resilience**. Addressing I&I also improves compliance with regulatory standards, reduces the risk of environmental incidents, and enhances the long term sustainability of sewer infrastructure.

To achieve these outcomes, utilities must combine reliable high resolution data with robust analysis methods, stakeholder engagement, and phased remediation programs that target the most impactful areas first. This integrated approach transforms I&I reduction from a reactive task into a proactive strategy for network management.

EDS brings over 30 years of experience in designing and delivering flow monitoring programs that transform I&I data into meaningful decisions. Our team partners with clients to build tailored solutions, from short term investigative studies to long term monitoring programs, helping optimise budgets, extend asset life and deliver measurable improvements in network performance.

About EDS

Environmental Data Services (EDS) is Australia's trusted partner in sewer flow monitoring and data driven network management. Our experienced field team, cutting edge equipment and ISO compliant data validation processes deliver the reliable insights needed for smarter infrastructure decisions.

Environmental Data Services

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